

Project Design Phase — II

Data Flow Diagram & User Stories

Date	24 July 2026
Team ID	Individual / Solo Project
Project Name	Software Installation Request Automation using ServiceNow Service Catalog
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Maximum Marks	4 Marks

Data Flow Diagrams (DFD)

A Data Flow Diagram is a traditional visual representation of information flows within a system. It shows how data enters and leaves the system, what processes transform the information, and where data is stored. For the Software Installation Request Automation project, a DFD helps clarify the flow of request data, approval decisions, license information, and deployment commands across ServiceNow, SAM, and endpoint management systems.

System Overview

The system receives software requests from employees via the Service Catalog, routes them through Flow Designer for approval decisions (checking with SAM for available licenses), updates license records, and triggers deployment through IntegrationHub to Intune/SCCM. Below is a simplified Level 0 DFD showing the major data flows and processes:

Data Stores	Processes

Key entities include the Employee (actor), Service Portal (interface), Flow Designer (decision logic), SAM (license validation), IntegrationHub (deployment orchestration), and SCCM/Intune (endpoint management). Data flows include the original request, approval decisions, license availability, deployment commands, and confirmation callbacks.

User Stories

User stories capture the functional requirements from the perspective of each stakeholder role. They follow the template: "As a [role], I want to [action] so that [benefit]." Below are the key user stories for the Software Installation Request Automation system:

User Type	Epic	US ID	User Story	Priority	Release
Employee	Catalog Discovery	EMP-1	As an employee, I want to find the Software Catalog item on the Service Portal so I can request the software I need.	High	MVP
Employee	Catalog Request	EMP-2	As an employee, I want to fill out a request form (software name, version, business justification) and submit it so I can formally request the software.	High	MVP
Employee	Status Tracking	EMP-3	As an employee, I want to see the real-time status of my software request (pending, approved, deploying, completed) so I know where my request stands.	High	MVP
Employee	Notifications	EMP-4	As an employee, I want to receive email and/or portal notifications when my request is approved, rejected, or ready to use so I don't have to keep checking manually.	High	MVP
Manager	Approval	MGR-1	As a manager, I want to receive approval requests in my worklist so I can review and approve/reject software requests from my team.	High	MVP

Manager	Approval Context	MGR-2	As a manager, I want to see the employee name, requested software, and business justification on the approval request so I can make an informed decision.	High	MVP
Manager	Escalation	MGR-3	As a manager, if I do not approve/reject within 3 days, the request should escalate to my skip-level manager so critical requests aren't blocked.	Medium	Phase-2
IT Admin	Fulfillment	IT-1	As an IT admin, I want approved requests to automatically trigger a fulfillment task and deploy the software via Intune/SCCM so I don't have to manually install.	High	MVP
IT Admin	SAM Integration	IT-2	As an IT admin, I want the system to check SAM for available licenses and automatically allocate one when an approval happens so we never exceed our license count.	High	Phase-2
IT Admin	Failure Handling	IT-3	As an IT admin, if a software deployment fails, I want to be notified immediately with error details so I can troubleshoot and retry.	High	MVP

These user stories form the foundation of the design and development phases. They ensure that the solution addresses the needs of each stakeholder and deliver value across the employee request experience, manager approval workflow, and IT fulfillment automation.